Cross Israel Highway Stormwater Quality Study
PHASE VI FINAL REPORT
SIMPTM Verification to 2009-2010 Collected Data

PROJECT REVIEW

Before the discussion of the Phase V & Phase VI results can be provided a review of the history of the entire project is in order. A 2.4 kilometer segment of Section 13 of the Cross Israel Highway (CIH), called for the purposes of this report, the Rosh Ha'ayin segment, drains upstream in the vicinity of the extremely vital Hayarkon Springs. In the spring of 2000, several parties raised a concern that runoff from this section of the CIH will impact the groundwater quality of the Hayarkon Springs. They petitioned the Supreme Court of Israel to issue an interim injunction against the continuation of the project until the issue of water contamination is resolved. The petitioners contended that runoff from the CIH, including toxic material spilled from traffic accidents, may pollute the underground water resources of the Hayarkon Springs. This section of the CIH drains to the Einat and Rosh Ha'ayin Channels. These two channels flow northwesterly and westerly respectively from the vicinity of the CIH. The Rosh Ha'ayin Channel merges with the Einat Channel immediately adjacent to and north of Highway 483. The Einat Channel then passes adjacent to the well fields of the Hayarkon Springs after which it flows into the Hyarkon River.

A group of decision-makers referred to as the Accompanying Team (AT) were appointed by the National Planning and Building Board (NPBB) to oversee the environmental aspects of the CIH and deal with the problem. The responsibility of the AT was to develop a plan of action for the CIH to address these concerns and then oversee the implementation of the plan. The Team developed a plan of action for the Rosh Ha'ayin segment referred to as the "Integrated Solution." Components relative to this study include the following:

- Design, construction and operation of spill containment facilities that will capture and contain any toxic spills that could occur along the this section of the CIH that drains to the vicinity of the Hayarkon Springs.
- Design and implement of a monitoring plan that obtains the data needed to eventually address the various concerns of the Accompanying Team.
- Conduct a stormwater modeling study that: characterizes the stormwater runoff from this section; determines the specific pollutant reductions associated with roadway cleanings; and provides a recommendation on the type and cleaning frequency of the most appropriate highway cleaning practice, if needed.

The Supreme Court ruled in November 2001 that the construction of the CIH will not be halted and the "Integrated Solution" will be implemented. The required spill containment facilities were constructed before the CIH opened to traffic in the fall of 2002. Each of the ten containment facilities have a control gate that remains closed at all times to contain potential spills. The facilities are designed to contain a total spill from the largest tanker truck known to transport hazardous liquids in Israel.

The CIH Monitoring Plan was developed by PWR/RPA and published a report in September 2003. The stormwater modeling study called the Cross Israel Highway Stormwater Quality Study was undertaken by PWR/RPA using a model called SIMPTM and published a final report in August 2004. Following the completion of the Final Report, the AT agreed to accept the report's recommendations that have been repeated below.

RECOMMENDATIONS ACCEPTED BY THE AT

1. Absent a national or watershed policy that leads to the reduction of non-point pollution from all sources in the watershed of the Einat and Rosh Ha'ayin channels, it is recommended that the Rosh Ha'ayin segment of the CIH not be cleaned or swept at this time.

2. To verify the results of the study, it is recommended that the CIH Ltd conduct further monitoring when the traffic volume reaches approximately 100,000 vehicles per day. The types of data collected should be specific to the needs of verifying the model simulations presented in this report. Current measures to protect against the potential harmful effects of accidental hazardous chemical spills should be continued.

The purpose of this report is to present the monitoring data collected from October 2009 though May 2010 and to check the accuracy of the SIMPTM simulations and validity of the conclusions that were reached at the end of Phase IV in August 2004.

INTRODUCTION TO PHASES V & VI

CIH, Ltd contacted PWR in August 2009 and instructed PWR to prepare a detailed monitoring plan for the collection of additional data that PWR felt was needed to verify the simulations of SIMPTM. PWR was also asked to develop a detailed scope of work and cost estimate for PWR's services needed in the completion of Phase V - Implementation of the Additional Monitoring Program and Phase VI- Verify Modeling Results and Prepare Documentation. The Phase V – Additional Monitoring Report was submitted to CIH, Ltd in draft form in September 2009. The final scope and budget of these phases were agreed to in November 2009.

The purpose of Phase V was to implement a program designed to collect the data necessary for the verification of the results presented in the 2004 PHASE IV FINAL REPORT. In that phase, the SIMPTM stormwater quality model was used to project the estimated storm-by-storm average annual washoff loadings and event mean concentrations for key pollutants. These pollutant loads were from the two different CIH pavement types (i.e. Traditional Asphalt and Partially Porous Asphalt) under the assumption that the highway traffic volume was equal to or exceeded 100,000 vehicles per day. In 2008, the highway traffic volume reached 140,000 vehicles per day, however, there was a highway-widening project under construction that added a third travel lane throughout this section of the CIH. As a result, the implementation of the additional monitoring program (i.e. Phase V) could not occur until the construction had been completed and any adjoining landscape areas were stabilized so the threat of eroded sediments entering the adjoining pavement had been eliminated. This non-eroding stable highway condition was reached in October 2009 when the additional data monitoring began.

POLLUTANTS OF INTEREST

The pollutants of interest in Phase V & IV remained the same as those identified in the Phase IV Report. They are as follows:

- Total Suspended Solids (TSS)
- Total recoverable chromium (TCr)
- Total recoverable copper (TCu)
- Total recoverable lead (TPb)
- Total recoverable zinc (TZn)
- Total nitrogen (TN)
- Total phosphorus (TP)

TSS was previously selected because it is the primary pollutant simulated by SIMPTM. Other pollutants are related to this simulation. The four metals in the above list were previously selected based on the review of the stormwater data collected during Phase II. This Phase II review suggested that exceedances of either the drinking water or draft river water standards may occur at a traffic volume of 100,000 vehicles per day. However, the monitoring data indicated that at the low traffic volume of about 20,000 vehicles per day during the initial monitoring period of October 2003 through April 2004 only the chromium drinking water standard was close to being exceeded.

Phase IV pointed out that since the concern regarding the CIH pavement discharges was directly related to the potential contamination of the drinking water supply of the Hayarkon Springs, then it was most appropriate to have the Accompanying Team (AT) focus on the Israeli Drinking Water Standards and the pollutants liable to exceed them. This left chromium concentrations above 0.05 mg/l as the only real water quality concern for the future high volume pavement discharges.

As a result, the Phase VI model verification modeling which this report presents focuses on the verification of the SIMPTM's previous projections of chromium concentrations from pavement discharges during high-traffic conditions in relationship to the Israeli drinking water standard of 0.05 mg/l.

PHASE V ADDITIONAL MONITORING PROGRAM

In October 2009, Mr. Roger Sutherland, P.E. of PWR traveled to Tel Aviv, Israel where he met with Mr. Reuven Levon of CIH, Ltd, consultant Prof. Uri Marinov , and both Mr. Nissim Keshet and Mr. Ariel Cohen of the National Parks Authority (NPA) to begin the implementation of the Phase V Monitoring Program. NPA was contracted by CIH, Ltd to collect the data that was needed as was the case previously in 2002-2003.

The objectives of the Phase V Monitoring Plan were as follows:

- Obtain rainfall depths for all storms that occur during the 2009-2010 wet weather season recorded by others at nearby gages.

- Determine the periodic quantity (mass per unit length of barrier) of road dirt accumulation found on both porous asphalt and traditional asphalt pavements located throughout the study section of the CIH.

- Determine the particle size distribution (PSD) of periodically sampled accumulated road dirt found on both porous asphalt and traditional asphalt pavements located throughout this section of the CIH.

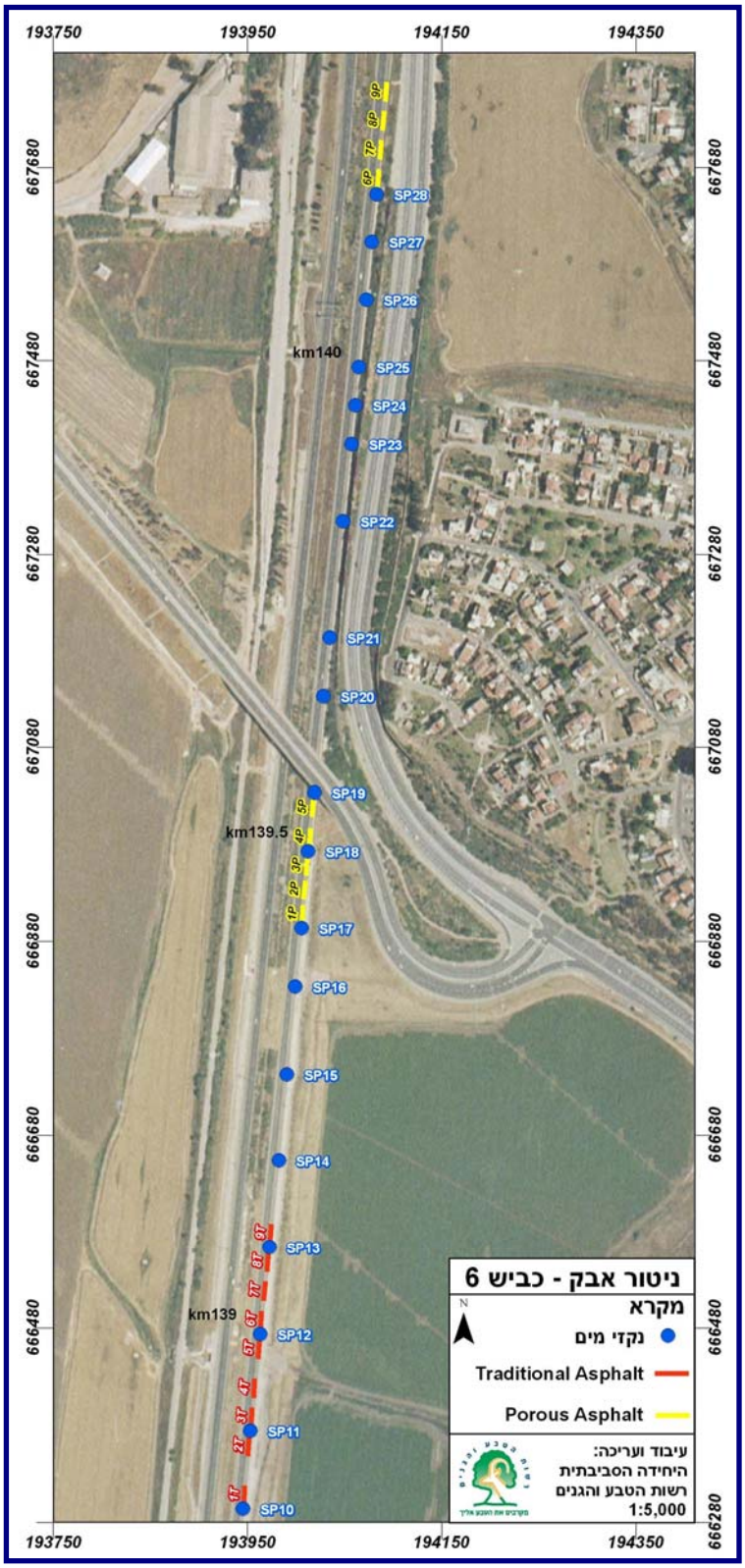
- Determine the key pollutant potencies (mg pollutant per kg road dirt mass) associated with the periodically sampled accumulated road dirt found on both porous asphalt and traditional asphalt pavements located throughout this section of the CIH.

ROAD DIRT ACCUMULATION SAMPLING

As outlined in the plan, Mr. Sutherland worked directly with Mr. Ariel Cohen to obtain the road dirt accumulation sampling equipment including an industrial vacuum cleaner and a gas-powered electric generator needed to monitor the accumulation of road dirt throughout the 2009-2010 wet-weather season. They both selected the multiple accumulation sampling sites for both the traditional asphalt and the porous asphalt pavements. Mr Sutherland also trained Mr. Cohen in how to collect road dirt accumulation samples.

Nine traditional (T) pavement sampling sites (i.e. numbered from 1 to 9 and labeled with a T) and nine porous (P) pavement sampling sites (i.e. numbered from 1 to 9 and labeled with a P) were identified along the right hand barrier of the north bound lanes of the CIH. Figure 1 (located to the right) shows the general location of each of the road dirt sampling sites along the CIH in the vicinity of the Hwy 483 overpass. Each sampling site was 20 meters in length with a buffer length of 10 meters or more separating one site from the next. The sampling site width was only one meter starting at the barrier and measured outward.

Mr. Sutherland also oversaw the important highway cleaning of this section that occurred on October 20, 2009. It was imperative that the highway was thoroughly cleaned by the cleaning machine and the shallow spillway pools at the stormwater discharge points were cleaned via a hand vacuum before the start of the sampling. The first sample collection occurred on the evening of October 27, 2009 with the sampling of Sites 1T & 1P. All of the samplings occurred during the late evening and nighttime hours due to safety concerns related to much higher traffic volumes during the daytime. The second sampling occurred on November 5, 2009 with the sampling of Sites 2T & 2P. A total of nine sampling events occurred with the last one on May 11, 2010. Each sampling site was sampled only once.



ROAD DIRT ACCUMULATION DATA

Table 1 presents the observed weighs and particle size distributions (PSD) of the road dirt accumulation data that was collected from October 27, 2009 through May 11, 2010 from both traditional asphalt and porous pavements along the study section of the CIH.

Table 1 - Road Dirt Mass and Particle Size Distribution (PSD) Observed on Porous and Traditional Pavements

Sample	Date	Mass	Particle Size Distribution (um)								
			8	7	6	5	4	3	2	1	Percen
		(g / 20 m)	> 6300	2000-6300	1000-2000	590-1000	250-590	125-250	62-125	< 62	Error
Porous Pavement											
P1	27.10.2009	4,164	5.2%	12.4%	8.2%	9.2%	27.2%	21.7%	7.5%	8.2%	0.3%
P2	5.11.2009	9,533	0.0%	25.0%	30.8%	14.9%	16.3%	8.4%	2.8%	1.7%	0.2%
P3	22.11.2009	10,017	1.0%	23.0%	26.6%	12.9%	19.8%	12.2%	2.6%	1.7%	0.3%
P4	15.12.2009	12,576	1.7%	19.3%	18.1%	11.1%	19.5%	14.6%	4.7%	10.2%	0.9%
P5	28.12.2009	29,827	1.1%	11.5%	11.1%	7.9%	18.9%	19.7%	8.6%	20.9%	0.6%
P6	28.1.2010	9,520	2.2%	28.3%	24.3%	11.1%	18.5%	11.7%	2.6%	1.1%	0.2%
P7	3.3.2010	5,869	0.4%	28.2%	23.6%	9.8%	17.0%	14.7%	4.4%	1.7%	0.3%
P8	7.4.2010	17,986	0.7%	22.8%	14.1%	10.0%	23.1%	19.1%	5.6%	4.4%	0.4%
P9	11.5.2010	21,539	6.3%	26.1%	11.5%	7.8%	20.0%	17.7%	5.3%	4.9%	0.4%
Traditional Pavement											
T1	27.10.2009	1,550	5.8%	16.0%	13.4%	13.0%	26.2%	17.2%	4.1%	4.0%	0.4%
T2	5.11.2009	3,998	4.4%	8.3%	15.1%	16.6%	33.0%	17.8%	3.2%	1.2%	0.4%
T3	22.11.2009	3,792	2.6%	17.2%	22.9%	16.8%	23.2%	12.4%	2.9%	1.9%	0.1%
T4	15.12.2009	4,118	3.0%	17.6%	17.8%	14.9%	25.4%	14.6%	4.1%	2.5%	0.1%
T5	28.12.2009	4,414	2.3%	17.3%	19.8%	15.1%	25.6%	14.3%	3.3%	2.1%	0.3%
T6	28.1.2010	2,990	0.8%	12.4%	15.6%	14.1%	28.8%	21.1%	4.5%	2.4%	0.3%
T7	3.3.2010	2,420	0.9%	8.7%	13.1%	14.5%	33.9%	23.3%	3.8%	1.5%	0.3%
T8	7.4.2010	4,392	2.3%	26.4%	16.5%	13.8%	24.0%	11.5%	2.9%	2.5%	0.2%
T9	11.5.2010	4,681	2.0%	33.7%	17.0%	12.7%	20.6%	10.3%	2.0%	1.4%	0.4%

It should be noted that the road dirt accumulations observed on the porous asphalt pavements are much greater than those on the traditional asphalt. This was the case in the previous monitoring collected during Phase II. This higher porous pavement accumulation is primarily due to the gutter channel (i.e.30 cms wide) that is created when the 50 mm porous asphalt pavement overlay is applied. The shallow channel that is created can easily trap and hold airborne road dirt deposited by passing traffic and windstorms. This gutter channel does not exist for the traditional asphalt pavements. The net result is much higher accumulations of road dirt being observed along the porous pavement sections when compared to traditional pavements. In fact 3.7 times more road dirt accumulated on porous pavements when compared to traditional. This ratio of porous accumulation divided by traditional accumulation is much greater than the 1.75 that was observed on high traffic volume highways in 2003 where porous pavement sections of Hwy 4 were monitored along with the traditional pavement sections of Hwy 1.

Figure 2 shows the range of PSDs that were observed from the 18 road dirt samples collected from either porous or traditional pavements. Table 2 presents the road dirt qualities that were observed for the pollutants of interest from the samples taken from both porous and traditional pavements. These qualities vary by the particle size range of the road dirt. The three ranges tested were the fine (F) range of less than 63 microns, the medium (M) range of 63 to 250 microns, and the course (C) range of 250 to 2000 microns. Figure 3 resents a plot of the total chromium road dirt quality data tabulated in Table 2.

Figure 2 - Road Dirt Particle Size Distributions (PSDs) Observed on Porous and Traditional Pavements

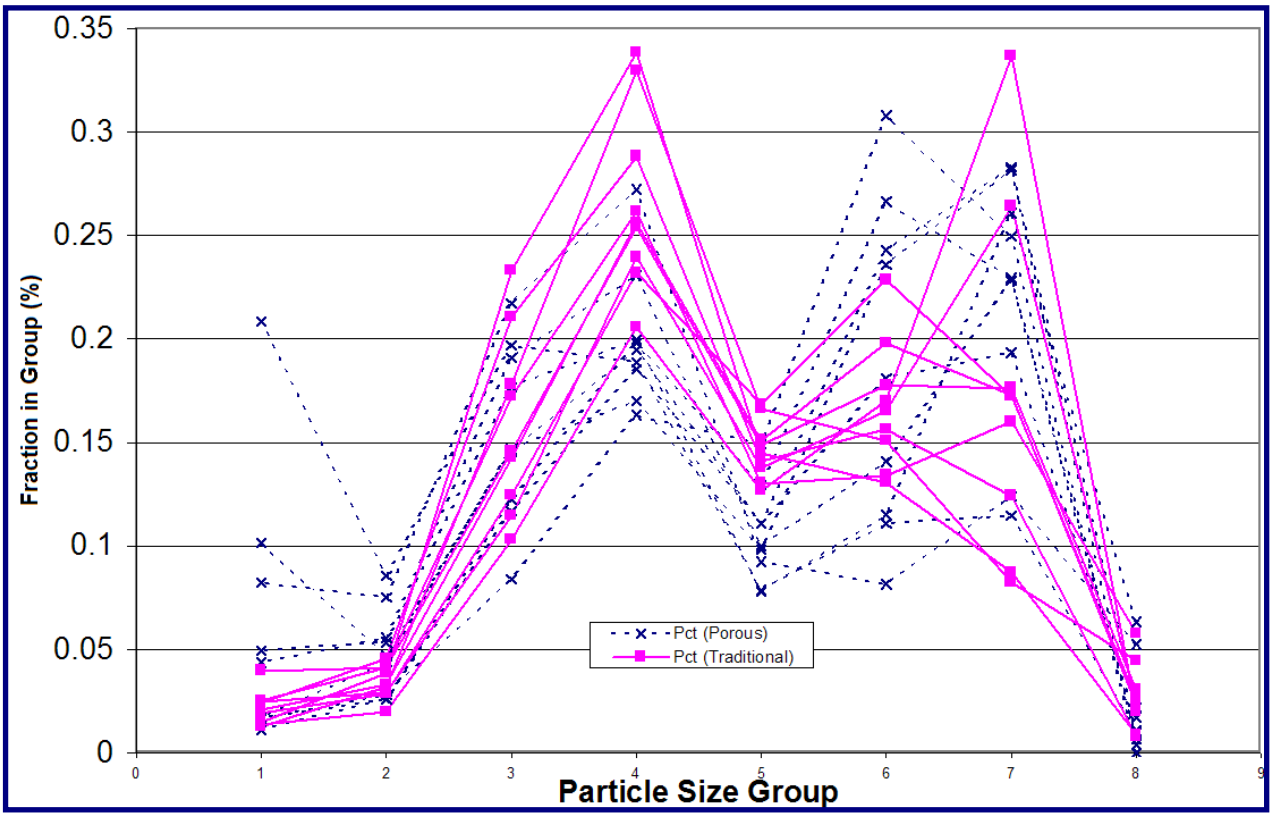


Table 2 - Road Dirt Qualities (mg/kg) for Pollutants of Interest Observed on Porous and Traditional Pavements

Sample	T-Cr			T-Cu			T-Zn			T-Pb			T-P			T-N		
	F	M	C	F	M	C	F	M	C	F	M	C	F	M	C	F	M	C
Porous Pavement																		
P1	90	37	45	50	14	43	250	90	60	73	30	14	300	≤ 200	≤ 200	1140	370	710
P2	120	46	65	64	18	21	200	140	50	75	63	10	≤ 200	≤ 200	≤ 200	500	200	160
P3	165	50	70	85	18	30	365	160	63	90	26	12	≤ 200	300	600	930	290	540
P4	87	35	60	42	24	20	224	115	40	46	43	15	400	300	400	600	300	240
P5	76	45	88	43	19	27	240	160	120	45	44	37	500	400	700	1000	410	380
P6	160	34	64	76	46	20	560	130	70	104	50	29	600	≤ 200	700	900	220	240
P7	116	47	116	65	38	28	415	150	65	80	35	28	300	≤ 200	900	520	< 120	150
P8	80	37	97	68	19	33	340	120	80	71	67	74	300	300	900	940	290	310
P9	85	85	100	57	12	26	350	100	74	60	42	52	400	400	1100	880	330	260
Traditional Pavement																		
T1	104	35	57	112	33	16	500	120	58	100	34	25	≤ 200	< 200	≤ 200	2280	470	930
T2	130	48	70	69	14	38	412	105	80	93	33	10	500	300	400	850	120	240
T3	172	50	75	80	17	18	385	142	64	230	56	47	600	≤ 200	300	940	130	160
T4	83	39	56	51	21	17	320	195	165	73	46	12	600	500	700	580	180	260
T5	87	40	87	60	46	17	340	250	60	66	20	12	500	≤ 200	300	1260	250	360
T6	92	55	90	66	14	24	335	150	200	77	44	23	600	300	400	670	140	160
T7	106	45	80	78	49	17	445	220	80	96	40	67	300	≤ 200	300	770	< 120	210
T8	96	42	53	76	89	18	300	183	95	92	70	11	500	≤ 200	400	730	180	420
T9	107	49	38	115	134	15	390	300	56	120	84	11	500	≤ 200	500	770	210	230

Figure 3 - Road Dirt Qualities for Total Chromium

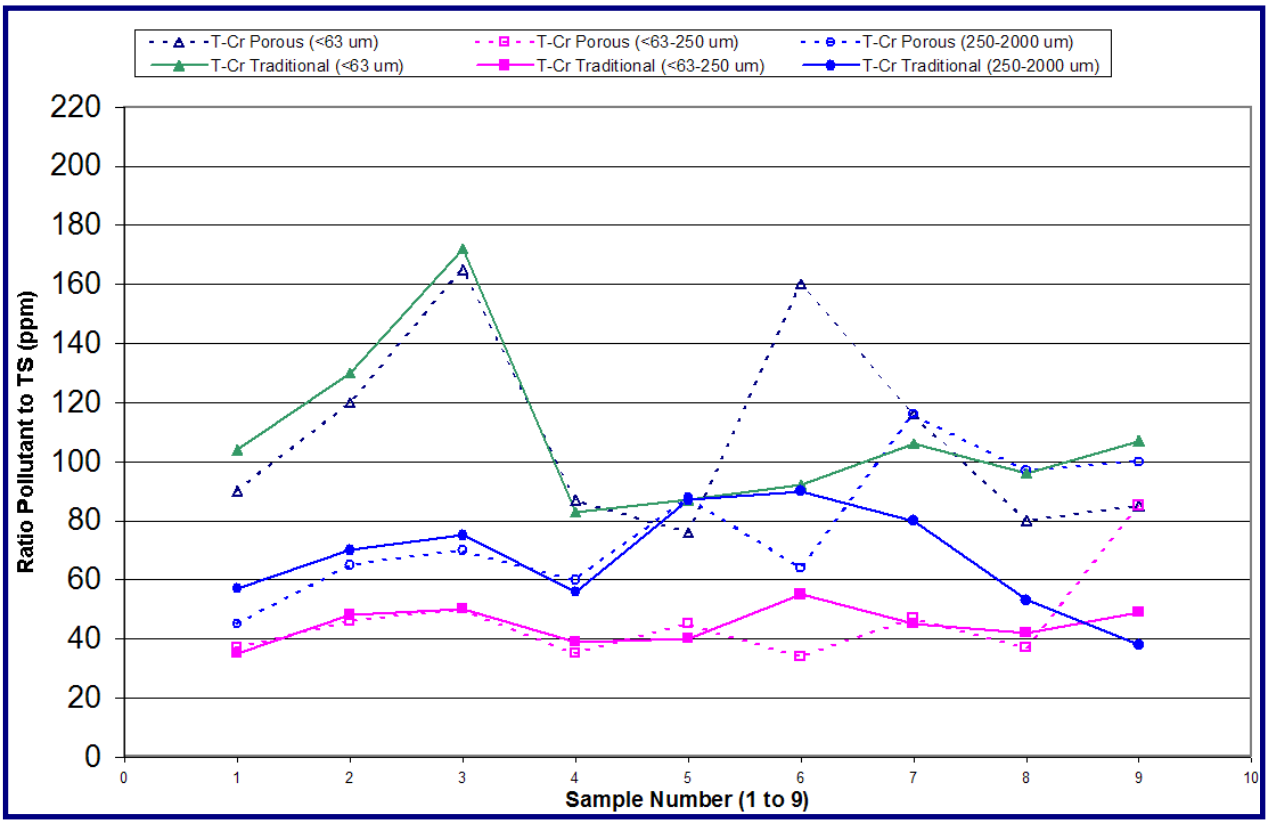


Figure 3 presents a plot of the total chromium road dirt quality data tabulated in Table 2. It is interesting to note that there was no chromium road quality data collected previously in Phase II. However, chromium was tested for in the runoff data that was collected during the Phase II wet season monitoring period of October 2002 through April 2003. The road qualities used for chromium in the Phase IV SIMPTM modeling were also calibrated to best reproduce these previously observed chromium discharge concentrations. The road dirt qualities used for total chromium were: 70 ppm for less than 63 microns; 80 ppm for 63-250 microns, and 10 ppm for 250-2000 microns. These total chromium road dirt qualities are somewhat different than those observed in the Table 2 data that averaged to: 109 ppm for less than 63 microns; 46 ppm for 63-250 microns; and 73 ppm for 250-2000 microns.

PHASE VI VERIFY MODELING RESULTS

Modeling provides an opportunity to simulate the site-specific pollutant accumulation and washoff processes that occur over time. By using a modeling approach, one can collect some data over a relatively short period of time on the important pollutant accumulation and washoff processes that occur. This information is used to calibrate the model, which is then used to simulate these accumulation and washoff behaviors over a much longer time frame of say several decades. The original goals for the modeling of stormwater quality from the Hayarkon Springs segment of the CIH were to determine: (1) whether pavement discharges would exceed water quality standards once the highway's ultimate traffic volume was realized, (2) if the answer was yes, then how frequently would these exceedances occur, and (3) how would these exceedances be reduce by implementing various alternative cleaning practices.

MODEL VERIFICATION PROCESS

Working with the observed the road dirt accumulations collected during the 2009-2010 sampling period and the rainfall data collected by others throughout the same period, the original intent of the Phase VI scope was to just rerun the final high traffic volume SIMPTM model used previously in Phase IV to estimate the these observed accumulations. A comparison of the observed verses simulated data for the 2009-2010 monitoring period would then determine how accurate the previous model actually was. So the intent was to not make any modifications to be model's previously calibrated parameters. Unfortunately, PWR overlooked that fact that a third driving lane was added to the study section so the previously calibrated parameter values would have to be adjusted anyways since the pavement area contributing runoff had increased over that assumed in 2004.

When PWR carefully examined the total pavement areas that were contributing runoff to each of the 18 sampling sites, it was determined that Porous pavement Sites 6P, 7P, 8P and 9P had very limited drainage areas with Sites 8P and 9P having the least. This means the amount of runoff volume available to wash off the accumulated road dirt along these porous pavement-sampling sites was significantly less than those in the previously modeled Phase IV porous pavement sites. Also Porous pavement Site 5P was located in a very flat area so the velocity of the runoff would not be as great as that of the previous Phase IV porous pavement sites. Both of these problems help explain why the observed road dirt accumulations on three of these five sites were so high (i.e. Sites 5P, 8P and 9P). For these reasons, SIMPTM parameter recalibration was needed to attempt to match the observed road dirt accumulation data collected during the monitoring period of October 2009 through May 2010. Rainfall events occurred throughout this period so matching the observed accumulation data would mean that the simulated washoffs that occurred during these pavement discharges were accurate also.

So the model verification process was adjusted to first recalibrate the SIMPTM model to closely reproduce the new road dirt accumulation data. Then the recalibrated model would use the same 57- year rainfall data set as used in Phase IV to simulate the long term total chromium event mean concentrations (EMCs) from both traditional and porous pavement sections. The number of simulated total chromium event concentrations that exceed the drinking water quality standard of 0.05 mg/l would then be determined.

RAINFALL DATA

For the Phase IV CIH Stormwater Quality Study, the long-term hourly rainfall record collected at the Ben Gurion International Airport, located some 10.6 km southwest of the study section was used. The Ben Gurion station provided a 57-year period of rainfall from January 1, 1938 through December 31, 1994 for the model simulations. This 57-year record of rainfall was used for the recalibrated SIMPTM model here in Phase VI also. However, the rainfall data used for the October 2009 through May, 2010 model recalibration period was collected at Beit Arye, some 12.5 km southeast of the study section. Unfortunately the Ben Gurion rainfall data for this time period was not available during the time frame that PWR did the model recalibration work in September 2010.

MODEL RECALIBRATION

Using the rainfall data from the Beit Arye gage, the SIMPTM model was recalibrated to best simulate the wet weather road dirt accumulation data previously presented. Figure 4 presents the road dirt accumulations and rainfall intensities observed throughout the monitoring period. Note the simulation started on October 20, 2009 (the date the pavements were thoroughly cleaned) and continued through May 11, 2010 which the date of the ninth and last sampling of the road dirt accumulations. The results of the recalibration for traditional pavements are shown in Figure 5.

Figure 4 - Road Dirt Accumulation and Rainfall Intensities Observed throughout the 2009-2010 Monitoring Period

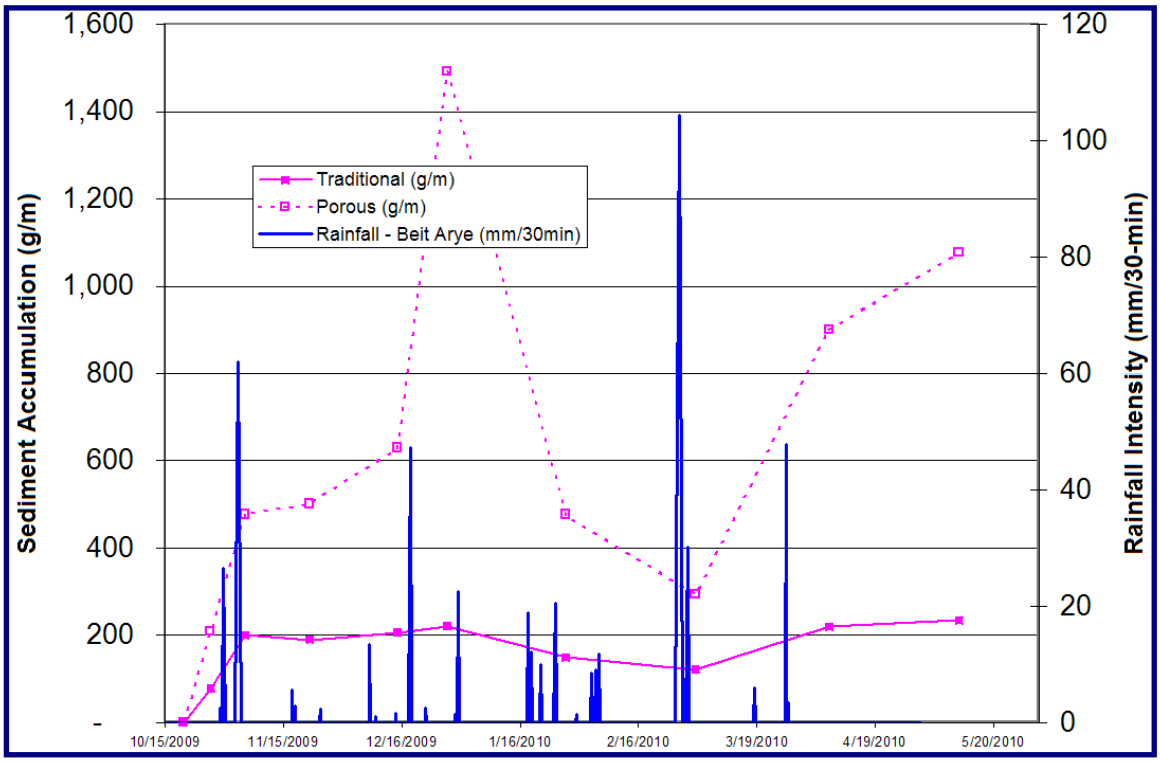


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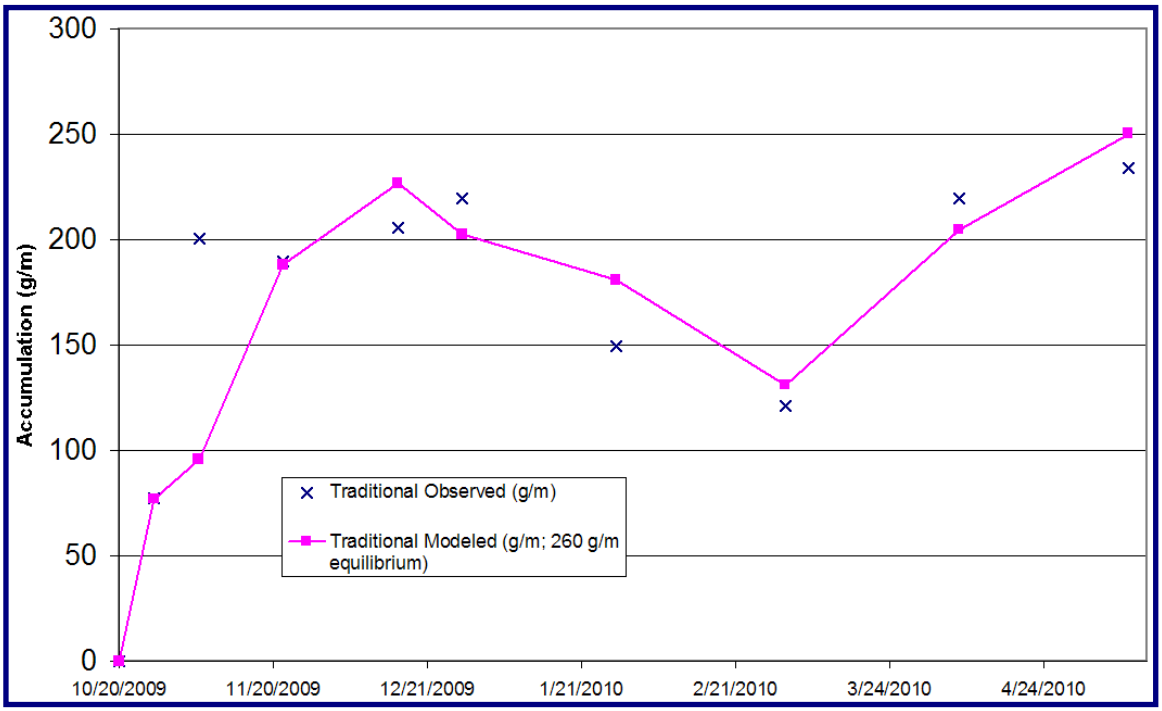
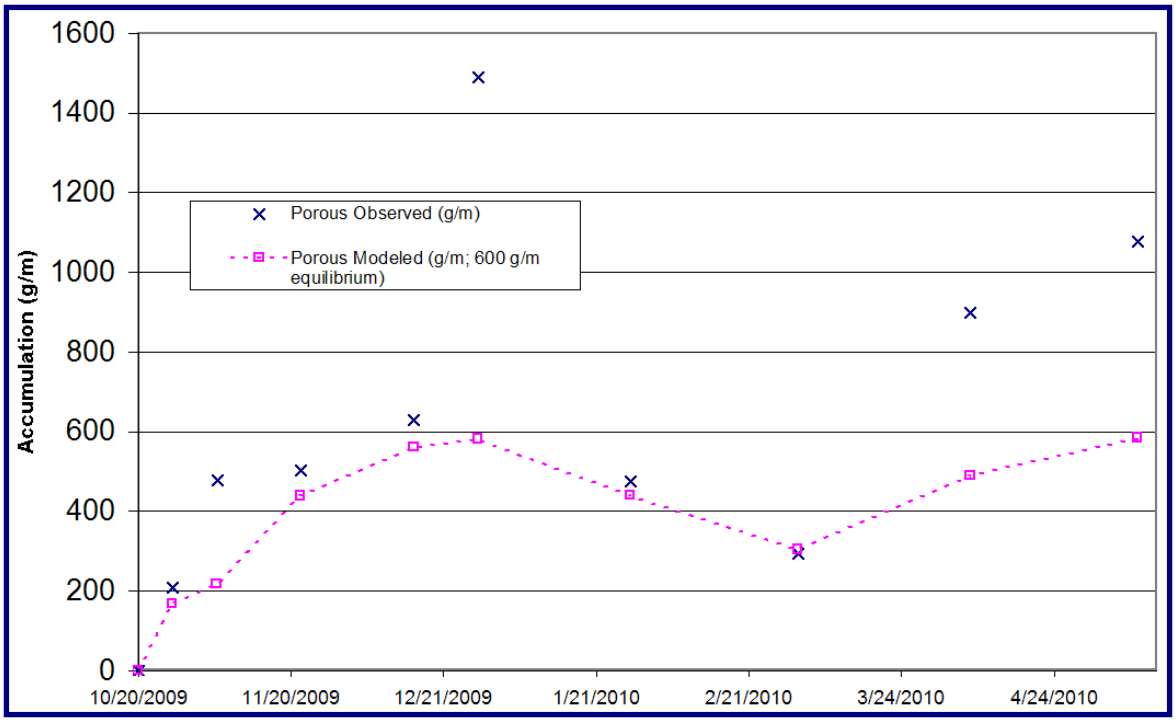
Figure 5 – SIMPTM Verification/Recalibration for Traditional Pavements



The model recalibration to the traditional pavement accumulations shown in Figure 5 was excellent. Two model estimates were very close to the values observed. Four model estimates were slightly over and two were slightly under. Site 2T was the only problem where the model estimate is significantly different being half the value that was observed. But this low estimate is still within the range of other observations since the Site 1T accumulation was lower. The Site 2T observation implies that the rate coefficient in the SIMPTM road dirt accumulation equation could be even greater than what was assumed in the recalibration. However, a higher rate coefficient failed to reproduce the excellent results shown in Figure 5 for the other 8 observations. Overall, it appears that modeled traditional pavement road dirt accumulations are about 50% higher than those projected in Phase IV for high traffic volumes.

Figure 6 – SIMPTM Verification/Recalibration for Porous Pavements



The model recalibration to the porous pavements shown in Figure 6 was fair. Model estimates for Sites 5P, 8P and 9P were significantly lower. But due to the problem previously described regarding the lack of drainage area to Sites 8P and 9P along with the essentially flat slope of Site 5P these three observations were though to be much higher than what should have been observed. Plus model estimate for Site 2P was also significantly lower. However this was a similar pattern to the traditional pavement Site 2T mentioned previously. Plus the Site 2P projected value seemed reasonable and within the range of the other good Site P observations. Overall it appears that modeled porous pavement road dirt accumulations are about 300% higher than those projected for high traffic volume in Phase IV.

LONG TERM MODEL SIMULATION RESULTS

The recalibrated SIMPTM model was used to simulate the storm-by-storm pollutant mass loadings and concentrations over the historic rainfall record under the current high traffic conditions with no cleaning operations. Table 3 presents the average annual pollutant event mean concentrations (EMCs) for stormwater discharges from the Rosh Ha'ayin segment of the CIH over the 57-year simulation period. The table also shows a comparison to the previous 2004 Phase IV calibrated SIMPTM model predictions for the high traffic condition which is now the current traffic condition. The average annual pollutant

EMC's for the recalibrated SIMPTM model are approximately 27% to 66% lower than the previous EMC estimates published in August 2004.

Table 3 - Average Annual Pollutant Event Mean Concentration (EMC) for the CIH[a] Under Current (High Traffic Volume) Conditions with No Cleaning

Pollutant	Previous Estimate mg/l	Current Estimate mg/l	Concentrations % Difference from Previous	Drinking Standards mg/l	Draft River Standards mg/l
TSS	443	152	-66%	NS[b]	NS
T-Cr	0.032	0.0223	-30%	0.05	0.01
T-Cu	0.026	0.0189	-27%	1.40	0.05
T-Pb	0.052	0.018	-65%	0.10	0.01
T-Zn	0.121	0.088	-27%	5.00	1.00
T-N	1.43	0.49	-66%	NS	NS
T-P	0.274	0.094	-66%	NS	NS

a. Total 4.66 km highway length is included in the drainage with 56% traditional pavements and 44% porous pavements
b. NS means no standard

Figure 7 presents the paired TSS and total chromium event mean concentrations simulated by the recalibrated SIMPTM for the 2037 storm discharges throughout the 57-year simulation period for traditional pavement discharges under the current high traffic volume conditions and with no road cleaning.

Figure 7 – Simulated TSS and Total Chromium Event Mean Concentrations From Traditional Pavements with Current High Traffic

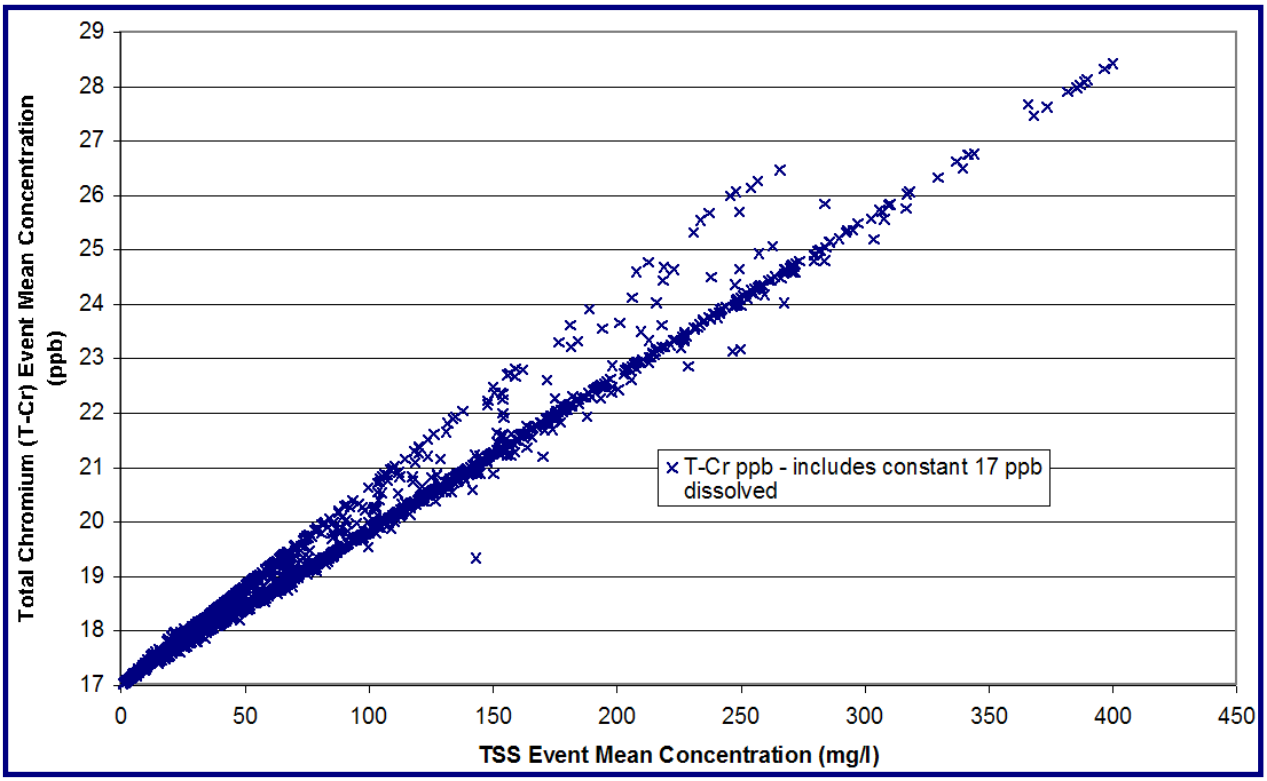
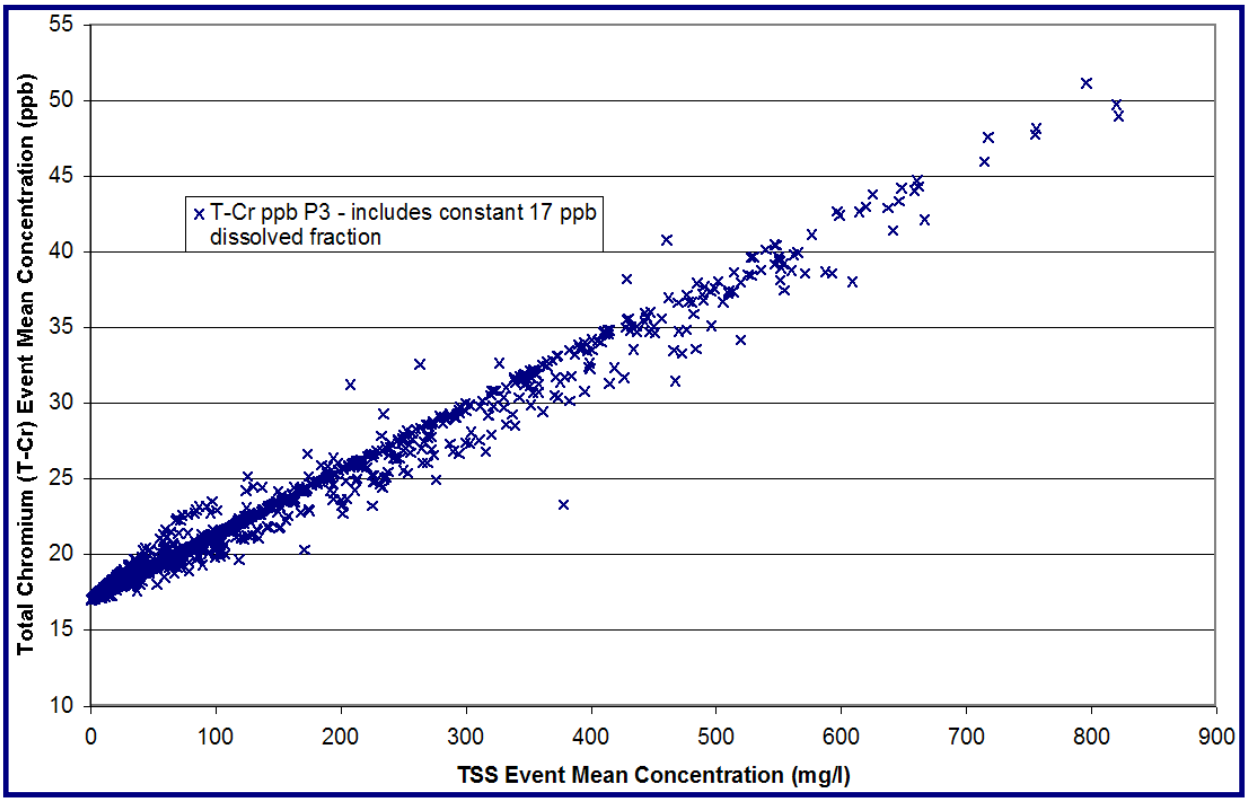


Figure 8 presents the paired TSS and total chromium event mean concentrations simulated by the recalibrated SIMPTM for the 2037 storm discharges throughout the 57-year simulation period for porous pavement discharges under the current high traffic volume conditions and with no road cleaning.

Figure 8 – Simulated TSS and Total Chromium Event Mean Concentrations From Porous Pavements with Current High Traffic



The EMCs presented in Table 3 for Total Chromium, Total Copper and Total Zinc reflect an appropriate dissolved metals correction that was identified in Phase IV and discussed in the 2004 Phase IV Report. Because of its continued importance here in Phase VI that discussion has been repeated below.

DISSOLVED METALS ADJUSTMENT

SIMPTM can simulate concentrations and loadings of Total Suspended Solids (TSS) found in urban stormwater. SIMPTM also simulates the particulate concentration and loadings of other pollutants such as metals based on the observed concentrations of those pollutants in the road dirt. However, SIMPTM does not directly simulate the dissolved fraction of each pollutant. SIMPTM is essentially a sediment particulate washoff model whose ability to simulate other pollutants is a function of the simulation of TSS washoff. Judgment was used, based on the stormwater quality data collected in 2002-2003 during Phase II, to add an appropriate value for the dissolved concentration for metals that exhibit a high dissolved fraction.

During the previous model calibration process in Phase IV, it was determined that the model satisfactorily simulated the particulate fraction of the various metals of interest. However, the model generally underestimated the total metals concentrations that were being observed at low TSS concentrations. This discrepancy occurs because the fraction of the dissolved pollutant was much greater at low TSS concentrations when compared to high TSS concentrations. To correct for the discrepancy, an adjustment was made to the particulate-based metals washoff simulations from SIMPTM. For chromium, zinc, and

copper, a single average dissolved metals concentration was added to the particulate concentrations simulated by SIMPTM. A dissolved metals adjustment for lead was not needed. Lead occurs primarily in the particulate state. Only 17% of the samples collected in Phase II were able to detect any dissolved lead. Whereas dissolved chromium, copper, and zinc were detected 96%, 98%, and 100% of the time respectively. The previous calibration for total lead was excellent without the need for an adjustment (see Figures E.13 and E.17 in Appendix E of the Phase IV Report).

Table 4 represents the statistical characteristics of dissolved metals data collected during the Phase II monitoring study. For each of the three metals in the table, the mean concentration shown was added to the particulate concentrations simulated by SIMPTM and presented earlier in Table 3 and Figures 7 & 8 to create total metal estimates.

Table 4 - Statistical Characteristics of Observed Concentrations of Dissolved Metals in CIH Pavement Runoff (from Phase IV Report)

Statistic	Dissolved Metal		
	Copper (mg/L)	Chromium (mg/L)	Zinc (mg/L)
Mean ($\bar{x}$)	.015	.017	.071
Median	.014	.011	.046
Standard Deviation (SD)	.004	.015	.065
Coefficient of Variation (COV)[a]	.282	.906	.913
70 Percentile[b]	.016	.019	.072
80 Percentile	.017	.026	.079
90 Percentile	.020	.040	.130

a COV = SD / $\bar{x}$
b Percentile is defined as the stated percentage of the observations in the dataset are less than or equal to the number shown.

DISSUSION OF PHASE VI MODELING RESULTS

The previous Phase IV findings put a focus on total chromium with respect to its potential exceedances of the Israeli drinking water quality standards. Total chromium was selected as the pollutant of greatest interest since the data collected in Phase II suggested that of the seven general pollutants of interest, total chromium was likely to have the highest frequency of exceedance of the Israel Drinking Water Standards once the high traffic volumes were realized.

However, Phase II monitoring results in 2002-2003 for direct pavement discharges from low traffic volumes on the CIH found no total chromium exceedances of the drinking water standard. Total chromium exceedances were observed for a few containment basin samples collected within 48 hours of a sampled storm event. Also very high concentrations of total chromium were found in both the Einat and Rosh Ha'ayin Channels both upstream and downstream of the CIH crossings.

Phase IV model simulations projected that total chromium concentrations from direct pavement discharges along the Rosh Ha'ayin segment under the conditions of approximately 100,000 vehicles per day traffic would exceed the Israeli drinking water standard of 0.05 mg/L or 50 ug/l for approximately 13% of the storm events. However, the results of the Phase VI model simulations from the recalibrated SIMPTM model presented above in Table 3 and Figures 7 & 8 clearly show that essentially all of the total chromium concentrations now predicted for direct pavement discharges from the Rosh Ha'ayin segment will remain below the 50 ug/l drinking water quality standard.

This surprising conclusion of no total chromium water quality violations seems counter intuitive since the road dirt accumulations observed in Phase V are obviously much higher than those previously projected

during the Phase IV modeling. The reason for this conclusion lies in the details of some of the original SIMPTM modeling parameter assumptions. Once it began clear that SIMPTM would have to be recalibrated to accurately reproduce the Phase V road dirt accumulation data since tributary drainage areas changed. These previous SIMPTM parameters were reexamined. Although a recalibration of the SIMPTM model was outside of the previously agreed too scope of services with CIH, Ltd., PWR felt it was in the best interest of the overall project, that began way back in 2000, to get the modeling results as accurate as possible.

Without getting into too many modeling details, the Phase VI SIMPTM parameter revisions related directly to the previous assumptions regarding the magnitude of the rainfall losses that occur when rainfall is transformed to stormwater runoff even though it lands on pavement. For example, in the case of partially porous pavements the runoff volume and timing of flow previously simulated in Phase IV did not reflect the actual conditions that could be expected to occur when rainfall lands and flows through 50 mm of porous asphalt overlaid on traditional asphalt pavement. Runoff peak flows would be greatly reduced which would have a very significant effect on reducing the amount of road dirt that actually washes out of the shallow gutter that was pavement design creates. When PWR revisited the washoff calibration graphs for both traditional and porous pavements presented in the Phase IV Report (Figure E.4 & E.5 of Appendix E), it was clear that the previous modeling parameter assumptions were greatly overestimating the amount of TSS being washed off the highway when compared to the values that were observed during the 2002-2003 wet weather monitoring period.

So the bottom line is that although road dirt accumulations observed in Phase V were much greater than those observed in Phase IV, the amount of the road dirt being transported by more realistic estimates of runoff gutter flows are much less that those previously estimate in Phase IV. This is seems to be the case based on the modeling results presented previously in Figures 5 & 6. Notice that a considerable amount of rain occurred throughout the Phase V monitoring period as shown on Figure 4. And several of these storms were very intense which is somewhat typical of the rainfall that does occur in the area. But the road dirt washoffs simulated for all of these rainfall events observed throughout the monitoring time frame had to be fairly accurate for SIMPTM to essentially estimate the actual road dirt accumulations that were observed. This is especially true for the traditional asphalt pavements that constitute the majority of the pavements in the Rosh Ha'ayin segment. By comparing Figures 5 & 6 from this recalibration to Figures E.4 & E.5 from the original Phase IV Report for the original calibration one can clearly see the recalibrated model is doing a much better job of accurately predicting the road dirt accumulation and wash off.

FINAL CONCLUSIONS

1. Traditional pavement road dirt accumulations observed in 2009-2010 along the Rosh Ha'ayn segment of the CIH are about 50% higher than those observed along Highway 1 in 2002-2003 and those projected by SIMPTM in 2004 for traffic volumes in excess of 100,000 vehicles per day which is the current traffic volume on the CIH.

2. Porous pavement road dirt accumulations observed in 2009-2010 along the Rosh Ha'ayn segment of the CIH are about 300% higher than those observed along Highway 4 in 2002-2003 and those projected by SIMPTM in 2004 for traffic volumes in excess of 100,000 vehicles per day.

3. Despite these greater that expected road dirt loadings, the long-term average annual pollutant event concentrations (EMC's) for the seven key pollutants of interest established in 2004 during Phase IV are approximately 27% to 66% lower than those previously estimated by SIMPTM for the high traffic condition.

4. The model simulations from the recalibrated SIMPTM model adjusted to include dissolved forms of chromium (as was done previously during Phase IV) clearly show that essentially all of the 2037 total chromium EMC's now predicted for direct pavement discharges from the Rosh Ha'ayin segment over a 57-year simulation period will remain below the 50 ug/l Israeli drinking water quality standard for total chromium.

5. Since the observed quality of water in 2002-2003 during Phase II for both the Einat and Rosh Ha'ayin channels upstream of the CIH crossings failed to meet the Israeli drinking water standards and the Rosh Ha'ayin segment represents only 1% of the watershed draining to the two channels at these crossing points, cleaning the CIH roadways within these watersheds will not provide any noticeable water quality benefits.

FINAL RECOMEDATIONS

- Absent a national or watershed policy that leads to the reduction of non-point pollution from all sources in the watershed of the Einat and Rosh Ha'ayin channels, it is recommended that the Rosh Ha'ayin segment of the CIH not be cleaned or swept at this time.

- Current measures (i.e. visual monitoring & eventual release of stored waters in the various containment basins) now being used to protect against the potential harmful effects of accidental hazardous chemical spills should be continued indefinitely.

- Since PWR has determined there is no longer any further risks to water quality now or in the future (other than the continued monitoring of any hazardous spill risk mentioned previously), the AT should accept this Phase VI Report as the final deliverable in a long list of deliverables and specific actions undertaken by CIH, Ltd to fully resolve the water quality concerns that were first identified in 2000 and associated with the construction of the Rosh Ha'ayin segment of the Cross Israel Highway

EPILOGUE

Pacific Water Resources, Inc. (i.e. Roger Sutherland & Seth Jelen) in association with Resources Planning Associates (i.e. Dr. Gary Minton) would like to personally thank Mr. Reuven Levon, Chief Engineer of CIH, Ltd, for the tremendous opportunity that he awarded to us. The trust that he placed in PWR back in October 2000 as the two firms in the entire world that could help him resolve these critically important local water quality concerns will never be forgotten. PWR/RPA accomplished this goal with Mr. Levon's support and the support of many of other CIH, Ltd staff members along with other dedicated Israeli professionals. Specifically, I would like to thank Mr. Nissim Keshet and Mr. Ariel Cohen and others at the Israeli Nature and National Parks Protection Authority for their tireless efforts in the vital collection of stormwater quality & road accumulation data during 2002-2003 and then again in 2009-2010. And special thanks to Professor Uri Marinov, consultant to CIH, Ltd, without his insight, knowledge, wit, encouragement, timely reviews and unending friendship this project would never have occurred or would never have been so successful. And finally, I want to thank the members of the Accompanying Team (AT) for their dedication and service to their country and their participation as the project's ultimate sounding board regarding the data collection, evaluations & analyses, modeling conclusions, and project recommendations. PWR/RPA wishes that the U.S. Federal Highway Administration (FHWA) and all of the state Department's of Transportation (DOT's) throughout the United States would take the potential risk that highways might pose to local surface water quality as serious as their Israeli allies clearly have.